

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> The power dissipated by the head light bulb is given by the formula $(P = I ^ 2 R \)$, where $(P \)$ is the power , $(I \)$ is the current , and $(R \)$ is the resistance . We need to find the current and resistance . The power $(P \)$ is 40 watts , and the voltage $(V \)$ across the bulb is 12 volts . So , $(I ^ 2 = \frac { P }{ R } = \frac { 40 }{ R } \)$ and $(I = \sqrt { \frac { 40 }{ R } } \)$.

Since the voltage across the bulb is 12 volts , we can use Oh m 's Law $(R = \frac { V }{ I } \)$. Rearrange for $(I \)$:

$[I = \frac { V }{ R } \]$

We need to find $(R \)$. Substitute $(I = \sqrt { \frac { 40 }{ R } } \)$ into $(R = \frac { V }{ I } \)$:

$[R = \frac { 12 }{ \sqrt { \frac { 40 }{ R } } } = \frac { 12 R }{ \sqrt { 40 } } = \frac { 12 R }{ 2 \sqrt { 10 } } \]$

$[R = \frac { 6 R }{ \sqrt { 10 } } \]$

$[R = 3 . 2 \text{ ohms } \]$

Now substitute $(R = 3 . 2 \)$ oh ms back into $(I = \sqrt { \frac { 40 }{ R } } \)$:

$[I = \sqrt { \frac { 40 }{ 3 . 2 } } = \sqrt { 12 . 5 } = 3 . 5 \text{ amps } \]$

Thus , the current is 3 . 5 amps and the resistance is 3 . 2 oh ms . </think>

< answer > J . 3 . 5 amps , 4 oh ms </answer>